

Szymon Galutowski

Portfolio | s.galutowski@gmail.com | linkedin.com/in/szymon | github.com/szymonator | British-Polish Citizen

EDUCATION

University of Warwick

Computer Science MEng

Coventry, England

Oct. 2025 – Present

Woodhouse College

Maths, Further Maths, Computer Science, Physics (all A-A*s)

London, England

Sep. 2023 – May 2025

EXPERIENCE

Research Intern

WMG, University of Warwick

Jul. 2026 – Sep. 2026

Coventry, England

- Engineered synthetic datasets using **GANs** to train and evaluate **Federated Learning** models.
- Analysed model weights, confusion matrices, and activation heatmaps across **MNIST**, **Fashion-MNIST**, and **SVHN** datasets.
- Identified variance in **L2 norms** between synthetic and real-world data to improve model security against adversarial attacks.

Data Annotator

Data Annotation

June 2025 – Sep. 2025

Westminster, London

- Engineered complex **JSON objects** to construct mock API calls, effectively stress-testing AI models to identify edge-case failure points.
- Evaluated and categorised AI responses to induce **reinforcement learning** and improve output accuracy.

Work Shadowing Intern

Wise & Surfboard

Jul. 2024

Westminster, London

- Shadowed multidisciplinary engineering and data teams across both companies, participating in **agile standups** and discussing technical problem-solving.
- Developed a Python script utilising **Pandas** to analyse and extract insights from financial trends.
- Built a web-based application demonstrating client-facing **API integration** and usage.
- Researched and presented on money transfer **security flaws**, proposing actionable architectural mitigations.

PROJECTS

Szymon's Movie Database | Java

- Engineered a high-performance **in-memory database engine** from scratch, achieving **sub-millisecond query execution** on a 1,000-movie dataset.
- Implemented custom data structures without standard libraries, including a dynamically resizing **hash map** and self-balancing **AVL trees**.
- Optimised range queries to **$O(\log n + k)$** time complexity and developed a generic **Quicksort** algorithm for reliable $O(n \log n)$ sorting.

Gesture Recognition Calculator | Python, JavaScript, MediaPipe, OpenCV

- Developed a touchless calculator interpreting physical hand gestures in real-time using **computer vision**.
- Curated and annotated a **custom dataset** of 1,000+ images to train **classification models** for digits and mathematical symbols.
- Ported the Python machine learning inference logic into **client-side JavaScript** for zero-backend browser deployment.

Warwick University Climbing Club Website | HTML/CSS, JavaScript

- Overhauled the official club website, removing proprietary **legacy software** to ensure long-term maintainability.
- Optimised asset delivery and implemented **semantic HTML** and **SEO best practices** to drive discoverability.

Check out my [portfolio](#) for much more!

TECHNICAL SKILLS

Languages: Python, Java, Haskell, JavaScript, SQL (Postgres), HTML/CSS, TypeScript

Frameworks: React, Node.js, Flask

Developer Tools: Git, VS Code, Visual Studio

Libraries / Tech: GANs, Federated Learning, Selenium, Pandas, NumPy, Matplotlib, OpenCV, MediaPipe